

Joint Sensing, Offloading, and Trajectory Optimization for Target Tracking in MLNTN

Rong Yuan, *Student Member, IEEE*, Shaohua Wu, *Senior Member, IEEE*, Chengjia Lei, and Qinyu Zhang, *Senior Member, IEEE*

Abstract—Non-terrestrial networks (NTNs) have emerged as a key component of global connectivity. However, single-layer unmanned aerial vehicle (UAV) or satellite solutions suffer outages, which remains a problem to support mission-critical services such as target tracking. To address mobility and jamming, we first design a multi-layer NTN (MLNTN) with clusters as operational units. Then essential factors influencing UAV swarms and satellite constellations are modeled, including relay routing and malicious interference. To address heterogeneous sensing between coarse satellite images and fine UAV images, we propose a collaborative detection approach with resolution preference, which flexibly locates targets from the acquired images, assisting in tracking. To enhance resource scheduling performance, we formulate a multi-objective problem as a decentralized partially observable Markov decision process (Dec-POMDP), jointly optimizing sensing selection, task offloading, and trajectory control. Furthermore, we introduce a joint-process-reward-shaping (JPRS) method to correlate task-oriented rewards and phase-oriented rewards. To realize distributed strategies, we propose two efficient multi-agent reinforcement learning (MARL) algorithms, facilitating a trade-off between execution timeliness and network resilience. Extensive experiments show the effectiveness of the proposed method, named multi-layer multi-agent proximal policy optimization (MLMAPPO), achieving a 22.15% reduction in time delay and a 10.21% improvement in mAP@0.5 relative to the PPO baseline, demonstrating the importance of collaborative detection for tracking in MLNTN.

Index Terms—Multi-agent reinforcement learning, non-terrestrial networks, target tracking, jamming, resolution.

I. INTRODUCTION

CURRENTLY, significant breakthroughs have been achieved in the collaborative strategy in non-terrestrial networks (NTNs), and the objective is to complete mission-critical services in a distributed pattern [1]. In this context, NTNs rely on the cooperation between space-based and air-based systems, and continue to provide more intelligent and autonomous services. On one hand, with the improvement of computation at the satellite-side, the transmission mode that only serves as bent-pipe communication has entered a bottleneck stage [2]. Thus orbit edge computing (OEC) has

emerged, which supports data processing onboard, greatly reducing the load and delay on the satellite-ground or satellite-UAV link [3]. In recent years, the applications of OEC have been extended to real-time earth observation [4], intelligent satellite node scheduling [5], and multi-objective optimization [6]. On the other hand, according to the Central Limit Theorem, the common impact of multi-jammers is considered Gaussian interference [7]. Due to closer proximity to jammers, a single unmanned aerial vehicle (UAV) is prone to malfunction by malicious interference, damaging communication with satellites. To enhance anti-jamming, a mobile ad-hoc network (MANET) in the form of UAV swarms has become a feasible solution [8]. With its unique flexibility, UAV swarms have successfully made significant contributions to solving practical problems such as emergency rescue and disaster relief, as well as inspection of power grids [9]. While most research has already analyzed satellite-UAV communication in NTNs, relay transmission represented by intra-cluster and inter-cluster communication is often overlooked. For example, by leveraging the relay mechanism of MANET, real-time information can be accessed between intra and inter UAV swarms, thereby enabling distributed target tracking [10]. Mega-constellations feature a long coverage time, but overhead time of a single satellite for ground users is relatively short. To realize sustained packet delivery, services in progress can be transmitted via the laser inter-satellite links (ISL) to the next satellite in the same or adjacent orbits [11]. Furthermore, even under differentiated resource conditions, end-to-end quality of service (QoS) can still be effectively supported [12].

As communication and computation become increasingly integrated within NTNs, incorporating sensing capabilities constitutes a pressing research challenge. To address this issue, a collaborative object detection framework based on edge-cloud is proposed in [13]. The device is responsible for lightweight target tracking, while object detection for region of interest (ROI) is completed on the server. Object detection is one of the key tasks in computer vision, whose core is to automatically identify and locate objects in images or videos as required. Compared with traditional methods relying on manually designed features, end-to-end deep learning methods demonstrate advantages of higher accuracy and real-time performance. Object detection has been extensively applied in terrestrial multi-access mobile edge computing (MEC) [14]. In recent years, multiple studies have explored object detection based on NTNs, mainly focusing on different types of task offloading [15], communication efficiency [16]. Despite making great progress, there remains a lack of consideration

Rong Yuan is with the Department of Electronics and Information Engineering, Harbin Institute of Technology (Shenzhen), Shenzhen 518055, China (e-mail: 24b952013@stu.hit.edu.cn).

Shaohua Wu and Qinyu Zhang are with the Guangdong Provincial Key Laboratory of Aerospace Communication and Networking Technology, Harbin Institute of Technology (Shenzhen), Shenzhen 518055, China, and also with the Peng Cheng Laboratory, Shenzhen 518055, China (e-mail: hitwush@hit.edu.cn; zqy@hit.edu.cn).

Chengjia Lei is with the Peng Cheng Laboratory, Shenzhen 518055, China, and also with the Department of Electronics and Information Engineering, Harbin Institute of Technology (Shenzhen), Shenzhen 518055, China (e-mail: 22b952020@stu.hit.edu.cn).

regarding edge computing and flying trajectory with resolution preference in NTN. For example, since UAVs are initially positioned far from targets, the images for object detection are only acquired by satellites. In contrast, images captured by UAVs from close range have higher accuracy as higher resolution. Although clear images make it easier to locate targets, this implies a rising cost of resources.

A. Challenge

Unlike terrestrial multi-access MEC environments, where nodes dynamically join or leave, a key distinction of NTN lies in using line-of-sight (LoS) for communication, and rapid movement across different regions renders it more complex. Specifically, NTN-based object detection faces the following key challenges:

- **Highly dynamic:** UAVs need to flexibly adjust trajectories according to flying situation, which leads to fast change in topology of MANET [17]. Due to periodicity of on-orbit, satellites have short overhead time and their beam coverage also fluctuates over time.
- **Malicious interference:** Malicious interference from the jammers impairs NTN, particularly slowing down target tracking and preventing UAVs from capturing high-resolution images. Relying solely on UAV swarms or satellite constellations, object detection will hardly achieve optimal regarding multi-objective problem.
- **Heterogeneous limitation:** Compared with the inherent blurring of satellite sensing, high resolution images captured by UAVs at close range can significantly improve tracking efficiency by accurate detection. However, due to limited onboard computational resources, local processing incurs considerable latency. One feasible approach is offloading the images to OEC, which unfortunately incurs high resource management overhead in space.

B. Motivation

Although existing studies on NTN have addressed resource management [18]–[20], these methods exhibit limitations when suffering outages (e.g., target tracking with jammers). To effectively address the above challenges, a hierarchical architecture needs to be developed, whose core advantages are summarized as twofold. (1) Clusters formed by UAV swarms and satellite constellations are adopted as the operational units. (2) A collaborative detection approach with resolution preference is adopted. Unlike vanilla NTN, all clusters should detect together with resolution selection to assist target tracking, thereby exhibiting superior resilience even under jamming. Leveraging the hierarchical architecture, edge nodes can be alleviated through multi-hop offloading.

C. Main Contributions

It is necessary to strike a balance between timeliness and resilience through full-process as well as closed-loop deployment of sensing, communication, computation, and trajectory. The main contributions of this paper are as follows:

- To address outage failures in dynamic topologies, we design a multi-layer NTN (MLNTN) with clusters as operational units. UAVs (e.g., Members, Leaders) and satellites enable intra-cluster, inter-cluster as well as satellite-UAV communication via multi-hop relaying. Ultimately,

a multi-objective optimization problem is formulated for sensing, offloading, computation, and trajectory.

- To address heterogeneous sensing within clusters, we propose a collaborative detection approach. This approach adaptively adjusts the resolution preference according to task requirements, significantly improving tracking efficiency. The multi-objective problem is transformed into a Dec-POMDP, and MARL techniques are employed. Moreover, a process-based reward-shaping method is proposed to balance individual tasks and system phases.
- To achieve joint optimization of time delay, energy consumption, outage robustness, and resolution preference, we propose two MARL algorithms for sensing selection, task offloading, and trajectory control. Convergence efficiency is evaluated against different MARL frameworks. Extensive experiments demonstrate that the MAPPO-based algorithm outperforms baselines, striking a trade-off between execution timeliness and network resilience.

D. Organization

The rest of this paper is organized as follows. Section II introduces the system model. Section III illustrates a novel approach for MLNTN-based object detection. Section IV formulates a multi-objective optimization problem. Section V presents the proposed MARL solution. We present experimental results in Section VI, and draw conclusions in Section VII.

II. RELATED WORKS

A. Resource Allocation in NTN

In remote areas lacking basic infrastructures, [18] utilizes a single UAV as a relay node to achieve seamless transmission of satellite signals across different orbits, aiming to forward packets between faraway ground terminals. [21] first proposes a multi-layer NTN but only involves satellite constellations to improve the energy efficiency of user terminals. However, these works all neglect to take UAV swarms into consideration, resulting in age of information (AoI) issues. It has been proven that NTN, which combine the complementary advantages of large-scale clusters formed by UAV swarms and satellite constellations, possess vast communication coverage and stable connection quality [22]. Furthermore, the deployment of object detection and MEC also plays crucial roles that cannot be ignored in resource management [23] [24]. To fully leverage space-based OEC, [19] proposes that after UAVs receive tasks from the ground, they offload to low Earth orbit (LEO) satellites for processing. Considering that UAVs are equipped with sensing devices (e.g., cameras) capable of capturing images with varying resolutions, [20] employs YOLOv5 for object detection and utilizes edge servers in ships for assisted computation. But this approach neglects resolution differences of sensing capability in NTN, lacking sufficient flexibility for services. Given the relay features of UAV swarms, [25] completes joint optimization of trajectories and communication in maritime search and rescue (SAR) by means of positioning with satellites and searching with UAV swarms. Similarly, when handling delay-sensitive tasks, transmission is conducted via UAV-LEO links, whereas for

delay-tolerant tasks, relaying can be realized through UAV-UAV links [26]. Moreover, interference is a special factor in the real world that cannot be ignored, generally making it difficult to provide service as usual. [27] considers a coordinated satellite-UAV communication system, where the UAV performs the environmental reconnaissance task with the assistance of satellites in hostile jamming. [28] analyzes anti-jamming strategies of task offloading in space-air-ground integrated network (SAGIN), providing a preliminary scheme in satellites-UAVs collaborative communication and computation for terrestrial edge nodes. Nevertheless, the aforementioned works still fail to fully conduct joint optimization of sensing, communication, computation, and trajectory under jamming.

B. AI-driven Techniques in NTN

Traditional heuristic methods faced with high computational complexity struggle to address mobility dynamics in NTN. With the rapid development of machine learning, reinforcement learning (RL) has emerged as a model-free and efficient approach for managing complex MEC networks [29]. It avoids the need for difficult non-convex transformations in multi-objective optimization and can approximate the optimal strategy. Specifically, it is considered to transform multi-objective problem into a Markov decision process (MDP), and design an adaptive strategy through intelligent technologies such as RL. MDP, consisting of five components: state space, action space, reward function, transition probability, and policy, is applicable to modeling key processes for SAGIN [30]. In [31], [32], RL algorithms such as Q-learning and proximal policy optimization (PPO) have been widely applied to resource scheduling in NTN, successfully addressing multi-objective MDP problems. When it comes to malicious interference, the passive defense paradigm, relying solely on the identification of interference, is gradually becoming outdated [33], [34]. An anti-jamming trajectory approach based on RL shows endogenous adaptability, leveraging inherent characteristics of agents [27]. Noting that if RL is applied to a multi-agent system (MAS), differentiated agents only obtain local observations, and the environment exhibits heterogeneity, typically resulting in suboptimal joint policies. Multi-agent reinforcement learning (MARL) [35], as an extension of RL, aims to alleviate challenges related to heterogeneous limitation and convergence in value estimation. In MARL, the multi-objective problem involving heterogeneous clusters is often formulated as a typical partially observable Markov decision process (POMDP) problem. [36] considers the centralized training and execution (CTE) architecture, which integrates multi-agent varying types into a fully centralized center for training. The drawback is that it tends to rely on fixed computing centers, making it hard to apply in remote areas [37]. To reduce dependency on the head quarter, a fully distributed framework represented by decentralized POMDP (Dec-POMDP), independent learning (IL) has been proposed, where each agent independently updates its policy based solely on its own observation and reward [38]. However, under IL deployment, some heterogeneous units may not pursue collaborative cooperation in order to maximize self-interest, or act as stragglers, delaying the overall convergence process [39]. Thus, the centralized training and decentralized execu-

tion (CTDE) paradigm has emerged, integrating the strengths and mitigating the weaknesses of both CTE and IL. In this paradigm, each agent shares information through centralized training and executes in decentralized ways, addressing heterogeneous limitation on numerous datasets [40].

In joint optimization problems related to MEC, MARL can fully achieve distributed operations, and its effectiveness has been proven [41]. Preliminary researches using MARL already have been conducted recently [42], yet there remain limited applications in NTN with clusters, especially satellite constellations. With the help of the CTDE paradigm, [43] introduces pigeon-inspired multi-agent deep deterministic policy gradient (MADDPG) algorithm to realize situation assessment and target assignment for UAV swarms in three-dimensional confrontation. In addition, [10] adopts the multi-agent proximal policy optimization (MAPPO) method with population-based learning, providing a dynamic task allocation scheme for UAV swarms executing SAR. Unlike these studies, we focus on a specific task: target tracking. The most relevant work to ours is [25], which utilizes heterogeneous vehicles for collaborative maritime SAR and employs a distributed MARL approach to optimize trajectory and communication processes of multi-target tracking, respectively. However, this study assumes that satellites only provide fixed positioning of multiple targets to guide UAVs tracking, failing to fully utilize the sensing capabilities as well as OEC in NTN. Thus, it is limited when handling different types of tasks, especially intensive computation. In contrast, we use MARL algorithms to design distributed policies of sensing selection, task offloading, and trajectory control while combining convex optimization to complete computation allocation. Moreover, our method leverages resolution preference between UAV swarms and satellite constellations, achieving collaborative detection in phases according to task requirements and tracking progress.

III. SYSTEM MODEL

We illustrate a hierarchical architecture named MLNTN targeted at remote areas in Figure. 1. All units in clusters accomplish detection and tracking of multi-target in a distributed manner. In this section, we introduce the system models, followed by malicious interference, mobility model, communication model, and outage robustness.

A. Malicious Interference

The effects of multiple non-periodic interference can be approximated as the following Gaussian distribution [7]. Malicious interference at different locations is determined by the power and channel gain of jammers [28].

We assume that there are Λ malicious jammers in total. The location of each malicious jammer in three-dimensional space is defined as $\mathbf{L}_k$, and its transmission power is P_k . For any mobile node $\mathbf{p}(t)$ in remote areas, the malicious interference from all jammers can be expressed as

$$I(t) = \sum_{k=1}^{N_s} P_k \times e^{(-\frac{(\mathbf{p}(t) - \mathbf{L}_k)^2}{2\sigma^2})}, \quad (1)$$

where σ is determined by Gaussian interference channel.

where $\beta_{j,s}(t)$ denotes the large-scale fading in L2S link. Similarly, it can be deduced as $\beta_{j,s}(t) = 10^{-\frac{PL_{j,s}(t)}{10}}$, and we can compute the path loss $PL_{j,s}(t)$ as

$$PL_{j,s}(t) = L_{\log}(t) - G_{j,s} + L_{space}(t), \quad (15)$$

where $L_{\log}(t)$, $G_{j,s}$ denote the logarithmic path loss like $PL_{j,j'}(t)$ and the total gain of antennas, respectively. Since the velocity of satellites (i.e., 7.5 km/s) is much higher, it results in a more pronounced Doppler shift. So, the frequency received by satellites $f_{s,j}$ can be derived as

$$f_{s,j} = (v_s/c) \times \left(\frac{R + z_j}{R + z_s} \cos \alpha_s \right) \times f_{j,s} + f_{j,s}, \quad (16)$$

where v_s is the velocity of satellites, α_s denotes azimuth angle, and R is radius of the Earth. z_j , z_s represent the altitude. $f_{j,s}$ denotes the transmission frequency of leaders and equals the carrier frequency f_{L2S} in L2S link. The antenna gains for sender and recipient can be written as [26]

$$g_j^{tx} = \phi \left(\frac{\pi f_{j,s} \Omega_j}{c} \right)^2, \quad g_s^{rx} = \phi \left(\frac{\pi f_{s,j} \Omega_s}{c} \right)^2, \quad (17)$$

where Ω_j and Ω_s denote the antenna radii. Due to signal attenuation caused by ionospheric ion absorption and solar magnetic storms during space transmission, this process can be represented as [48]

$$L_{space}(t) = 10^{\left(\frac{3\psi d_{j,s}(t)}{10z_s} \right)}, \quad (18)$$

where ψ represents attenuation in space. The transmission rate between access j and satellite s is calculated as

$$R_{j,s}(t) = B_{j,s} \log_2 \left(1 + \frac{P_{j,s} \text{Channel}_{j,s}(t)}{\sigma^2} \right), \quad (19)$$

where $P_{j,s}$ is the transmit power of access leader j , and the bandwidth for L2S link is $B_{j,s}$.

D. Outage Robustness

Relaying leaders may experience transient communication failures such as outage of distance, interference, and overhead [25]. For each member, the number of transmission paths is at most $\sum_{j=1}^J A_j^j$. Since each UAV cluster is guaranteed in coverage of satellites, the number of transmission paths enabled by L2S link is at most $(\sum_{j=1}^J A_j^j + \aleph)$. Thus, the outage robustness of MLNTN can be derived as

$$\Gamma(t) = \frac{1}{U} \sum_u \frac{\mathcal{F}_u(t)}{\sum_{j=1}^J A_j^j + \aleph}, \quad (20)$$

where A_j^j denotes a permutation operation based on factorials, and $\mathcal{F}_u(t)$ represents the number of established communication links for member u .

IV. COLLABORATIVE SENSING FOR MLNTN-BASED OBJECT DETECTION

To address heterogeneous sensing between UAVs and satellites, we propose a collaborative detection approach with resolution preference for MLNTN, aiming to improve tracking efficiency with minimal impact on workload.



Fig. 2. With deployment of object detection, UAV swarms or satellite constellations acquire images captured by onboard cameras. The bounding box is used to derive the coordinates of multi-target for tracking.

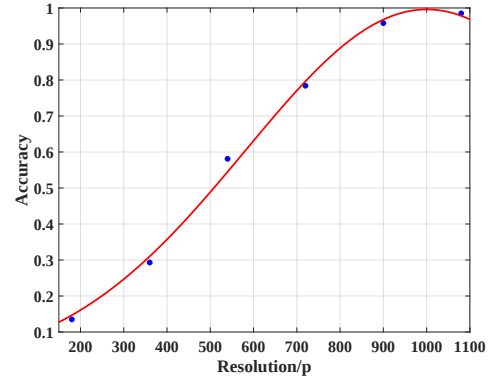


Fig. 3. Accuracy versus resolution in object detection.

A. Resolution Preference

MLNTN needs to conduct distributed tracking of multi-target based on detection as illustrated in Fig. 2. The set of optional resolutions is denoted as $G = \{\rho^1, \rho^2, \dots, \rho^g\}$. Specifically, G_{low} and G_{high} are defined as the coarse-grained set and fine-grained set respectively, with $G = G_{low} \cup G_{high}$. The image resolution $\rho_u(t) \in G$ denotes the sensing selection made by member u . YOLOv5s [49] is employed to obtain statistical correlation between accuracy and resolution through real ship-detection datasets [50]. If an object is successfully detected, it must simultaneously satisfy that the label of the bounding box is consistent with the ground truth and the intersection over union (IoU) of the bounding box is greater than the preset threshold. Specifically, coarse-grained images captured by satellites and fine-grained images captured by UAVs are cropped at different resolutions in preprocessing, respectively. Then, extensive testing is conducted using YOLOv5s, and curve fitting is performed with a Gaussian function, as the results shown in Fig. 3. It can be observed that the accuracy of object detection and image resolution exhibit a typical concave function characteristic [20].

In particular, considering UAVs are initially far from the targets, detection can only be achieved through coarse-grained images from satellites. Only after targets enter the coverage of UAVs can fine-grained images shot by UAVs be acquired or coarse-grained images continue to be selected for detection. Therefore, the key to improving the accuracy of object detection lies in acquiring high-resolution images selectively to reduce unnecessary resource overhead, and thus the resolution preference for MLNTN is defined as

$$\eta(t) = \frac{1}{U} \sum_u \text{if}(\rho_u(t) \in G_{high}). \quad (21)$$

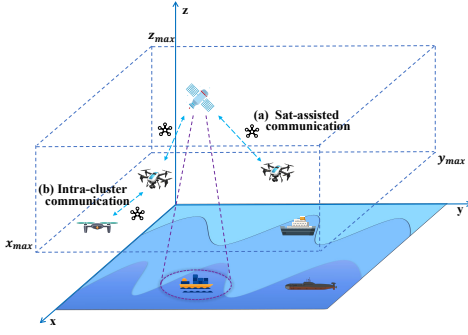


Fig. 4. Case of (a) one-hop and (b) multi-hop in the UAV routing network. Coarse-grained images would be relayed through sat-assisted or intra-cluster communication with the corresponding hop.

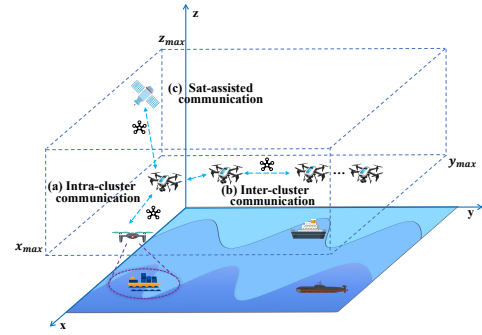


Fig. 5. Case of (a) one-hop and (b) (c) multi-hop in the UAV routing network. Fine-grained images would be relayed through intra-cluster, inter-cluster or sat-assisted communication with the corresponding hop.

B. Offloading Scheme

The tasks generated by object detection can be represented as $\text{Task}_u(t) = (d_u(t), c_u(t), T_u^{\max}(t))$, where $d_u(t)$ denotes the data size in bits, $c_u(t)$ denotes the computational density in CPU cycles per bit, and $T_u^{\max}(t)$ denotes the delay requirement for task execution in seconds, thus we have

$$d_u(t) = a\rho_u^2(t), \quad (22)$$

$$c_u(t) = b d_u(t), \quad (23)$$

where a and b are constants determined by the image format and detection network. The size of image to be detected and loaded for CPU processing is correlated with $\rho_u(t)$.

Let $x_u(t) = \{\alpha_u(t), \beta_{u,j}(t), \gamma_u(t)\}$ represent an offloading decision made by member u . Since tasks are indivisible and can only be processed at a single node, this corresponds to binary offloading. Unlike existing approaches, MLNTN enables collaborative detection between satellites and UAVs. Specifically, collaborative detection involves adaptive image acquisition and detection at the optimal node to localize the target, thereby assisting in tracking. Thus, the task offloading scheme can be classified into the following two cases.

(1) *Tasks are acquired by satellites:* As illustrated in Fig. 4, when $x_u(t) = 0$, satellite s offloads tasks to member u . Given that output data is extremely small compared to input, the transmission delay and energy are generally neglected when retrieving. f_u denotes the computational capability of member u , so we derive time delay and energy consumption as

$$T_u^M(t) = \frac{d_u(t)}{R_{s,j}(t)} + \frac{d_u(t)}{R_{j,u}(t)} + \frac{d_u(t)c_u(t)}{f_u} + t_{prop}, \quad (24)$$

$$E_u^M(t) = P_s \frac{d_u(t)}{R_{s,j}(t)} + P_{j,u} \frac{d_u(t)}{R_{j,u}(t)} + \kappa f_u^2 d_u(t) c_u(t), \quad (25)$$

where κ is the energy consumption coefficient, and t_{prop} is the propagation delay. If $\alpha_u(t) + \sum_{j=1}^J \beta_{u,j}(t) \neq 0$, satellite s offloads tasks to any leader j . The computational capability of leader j is $f_j(t)$, then we have

$$T_u^L(t) = \frac{d_u(t)}{R_{s,j}(t)} + \frac{d_u(t)c_u(t)}{f_j(t)} + \frac{2d_{u,j}(t)}{c}, \quad (26)$$

$$E_u^L(t) = P_s \frac{d_u(t)}{R_{s,j}(t)} + \kappa f_j^2(t) d_u(t) c_u(t). \quad (27)$$

When $\gamma_u(t) \neq 0$, satellite s computes tasks onboard. The computational capability of satellite s is $f_s(t)$, and we can get

$$T_u^S(t) = \frac{d_u(t)c_u(t)}{f_s(t)} + N \times \frac{d_u(t)}{Q_{\text{ISL}}}, \quad (28)$$

$$E_u^S(t) = \kappa f_s^2(t) d_u(t) c_u(t) + N \times P_s \cdot \frac{d_u(t)}{Q_{\text{ISL}}}, \quad (29)$$

where Q_{ISL} is the transmission rate of ISL, and N is the number of hops [11].

(2) *Tasks are acquired by UAVs:* As illustrated in Fig. 5, when $x_u(t) = 0$, member u computes tasks onboard, so this process can be written as

$$T_u^M(t) = \frac{d_u(t)c_u(t)}{f_u}, \quad (30)$$

$$E_u^M(t) = \kappa f_u^2 d_u(t) c_u(t). \quad (31)$$

When $\alpha_u(t) + \sum_{j=1}^J \beta_{u,j}(t) \neq 0$, member u offloads tasks to access leader j or helper leader j' . If member u offloads task to access leader j through intra-cluster communication, it can be written as

$$T_u^L(t) = \frac{d_u(t)}{R_{u,j}(t)} + \frac{d_u(t)c_u(t)}{f_j(t)} + \frac{2d_{u,j}(t)}{c}, \quad (32)$$

$$E_u^L(t) = P_{u,j} \frac{d_u(t)}{R_{u,j}(t)} + \kappa f_j^2(t) d_u(t) c_u(t). \quad (33)$$

If member u offloads tasks to helper leader j' through inter-cluster communication, which indicates the access leader j has been overloaded for processing object detection, so this process can be derived as

$$T_u^L(t) = \frac{d_u(t)}{R_{u,j}(t)} + \frac{d_u(t)c_u(t)}{f_j(t)} + \frac{2d_{u,j}(t)}{c} + T_{\text{multi}}, \quad (34)$$

$$E_u^L(t) = P_{u,j} \frac{d_u(t)}{R_{u,j}(t)} + \kappa f_j^2(t) d_u(t) c_u(t) + E_{\text{multi}}, \quad (35)$$

where T_{multi} , E_{multi} represent time delay and energy consumption within multi-hop routing, respectively.

When $\gamma_u(t) \neq 0$, member u offloads tasks to satellite s passing by access leader j , this process can be written as

$$T_u^S(t) = \frac{d_u(t)}{R_{u,j}(t)} + \frac{d_u(t)}{R_{j,s}(t)} + \frac{d_u(t)c_u(t)}{f_s(t)} + \frac{2(d_{u,j}(t) + d_{j,s}(t))}{c} + N \times \tau^{\text{trans}}, \quad (36)$$

$$E_u^S(t) = P_u \frac{d_u(t)}{R_{u,j}(t)} + P_{j,s} \frac{d_u(t)}{R_{j,s}(t)} + \kappa f_s^2(t) d_u(t) c_u(t) + N \times P_s \cdot \tau^{\text{trans}}. \quad (37)$$

Therefore, the total time delay and energy consumption for MLNTN can be expressed as

$$\begin{aligned} T(t) &= \frac{1}{U} \sum_{u=1}^U \left\{ \left[1 - \alpha_u(t) - \sum_{j=1}^J \beta_{u,j}(t) - \gamma_u(t) \right] T_u^M(t) \right. \\ &\quad \left. + \left[\alpha_u(t) + \sum_{j=1}^J \beta_{u,j}(t) \right] T_u^L(t) + \gamma_u(t) T_u^S(t) \right\}, \quad (38) \\ E(t) &= \frac{1}{U} \sum_{u=1}^U \left\{ \left[1 - \alpha_u(t) - \sum_{j=1}^J \beta_{u,j}(t) - \gamma_u(t) \right] E_u^M(t) \right. \\ &\quad \left. + \left[\alpha_u(t) + \sum_{j=1}^J \beta_{u,j}(t) \right] E_u^L(t) + \gamma_u(t) E_u^S(t) \right\}. \quad (39) \end{aligned}$$

V. PROBLEM FORMULATION AND TRANSFORMATION

A. Formulation

For efficient sensing, offloading, computation, as well as trajectory optimization, we aim to enhance timeliness and resilience. The multi-objective problem can be formulated as

$$\text{P1: } \min_{\rho, x, f, v} \frac{1}{T} \sum_{t=1}^T \lambda_1 T(t) + \lambda_2 E(t) - \lambda_3 \Gamma(t) - \lambda_4 \eta(t), \quad (40a)$$

$$\text{s.t. } \rho_u(t) \in G, \forall u \in \mathcal{U}, \forall t \in \mathcal{T}, \quad (40b)$$

$$\sum x_u(t) \leq 1, \forall u \in \mathcal{U}, \forall t \in \mathcal{T}, \quad (40c)$$

$$x_u(t) \in \{0, 1\}, \forall u \in \mathcal{U}, \forall t \in \mathcal{T}, \quad (40d)$$

$$T(t) \leq T_u^{\max}(t), \forall u \in \mathcal{U}, \forall t \in \mathcal{T}, \quad (40e)$$

$$\sum f(t) \leq f^{\max}, \forall j \in \mathcal{J}, \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, \quad (40f)$$

$$0 \leq f(t) \leq f^{\max}, \forall j \in \mathcal{J}, \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, \quad (40g)$$

$$v^{\min} \leq v(t) \leq v^{\max}, \forall u \in \mathcal{U}, \forall j \in \mathcal{J}, \forall t \in \mathcal{T}, \quad (40h)$$

where λ are weight factors used to measure the importance of various optimization objectives. ρ_u denotes sensing selection made by member u and belongs to total resolution set G . Similarly, x_u denotes task offloading also made by member u while satisfying binary offloading. $T_u^{\max}$ represents maximum time limit of tasks accomplished. f is decision of computation allocation mainly made by leaders or satellites, and it is necessary to ensure it does not exceed capacity. v denotes the flying velocity of UAVs, including speed and direction.

Due to the highly dynamic characteristics, the joint strategies pertain to a long-term stochastic optimization problem.

Upon observing the multi-objective function, it is obvious that P1 is a mixed-integer nonlinear programming (MINLP) problem with coupled variables. Meanwhile, the task offloading problem can be regarded as a Generalized Assignment Problem (GAP), which has been proven to be an NP-hard problem. Since P1 contains the GAP as a special case, P1 is NP-hard [39], which means that conventional methods find it extremely challenging to obtain the optimal solution.

B. Transformation

Above all, we decompose P1 into a continuous computation allocation sub-problem and a remaining discrete sub-problem. The former can be addressed on the side of CPUs using convex optimization [51], while the remaining sub-problem will be tackled through MADRL. For given decisions of sensing selection, task offloading, and trajectory control, the computation allocation sub-problem is written as

$$\text{P2: } \min_{\mathbf{f}} \frac{1}{T} \sum_{t=1}^T \lambda_1 T(t) + \lambda_2 E(t), \quad (41a)$$

$$\text{s.t. } T(t) \leq T_u^{\max}(t), \forall u \in \mathcal{U}, \forall t \in \mathcal{T}, \quad (41b)$$

$$\sum f(t) \leq f^{\max}, \forall j \in \mathcal{J}, \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, \quad (41c)$$

$$0 \leq f(t) \leq f^{\max}, \forall j \in \mathcal{J}, \forall s \in \mathcal{S}, \forall t \in \mathcal{T}. \quad (41d)$$

We decouple P2 as an instantaneous one within each time slot, and computation allocation at CPUs can be obtained as

$$\text{P3: } \min_{\mathbf{f}(t)} \lambda_1 T(t) + \lambda_2 E(t), \quad (42a)$$

$$\text{s.t. } T(t) \leq T_u^{\max}(t), \forall u \in \mathcal{U}, \quad (42b)$$

$$\sum f(t) \leq f^{\max}, \forall j \in \mathcal{J}, \forall s \in \mathcal{S}, \quad (42c)$$

$$0 \leq f(t) \leq f^{\max}, \forall j \in \mathcal{J}, \forall s \in \mathcal{S}. \quad (42d)$$

According to the properties of convexity preserving operations, it can be easily proven that the optimization objective and constraints with respect to $\mathbf{f}(t)$ are both convex, thus making the problem a convex optimization problem. Therefore, we can use a convex optimization toolbox such as CVXPY to obtain the optimal solution $\mathbf{f}^*(t)$.

C. Dec-POMDP for MLNTN

Due to heterogeneous limitation, the traditional algorithms that adopt a centralized structure are no longer suitable. In the following process, MLNTN is described as a Dec-POMDP problem to solve, which is mainly composed of $\langle \mathcal{O}, \mathcal{S}, \pi, \mathbf{r} \rangle$. Among them, $\mathcal{O}$ indicates the observable information, $\mathcal{S}$ indicates the state space, π indicates the action space, and $\mathbf{r}$ indicates the reward function.

1) *Observation*: Considering the system operates with a distributed pattern, this implies non-stationarity for each agent. The information used for decision-making is based on local observation, including own coordinates, action policy, as well as relative positions of targets, jammers, and other UAVs. It is worth noting that some slight differences exist between agents. For example, since the goal of the leader is to keep

connectivity stable, there is no need to contain the position of targets. The observation tuple is defined as

$$\mathcal{O}(t) = \{\text{relative pos, cor, } \pi\}. \quad (43)$$

2) *State*: Generally, the state is formed by stacking the observation information of all agents and eliminating redundancies. Herein, this process can be defined as

$$\mathcal{S}(t) = \{(\bigcup_{u \in \mathcal{U}} \mathcal{O}_u(t)) \bigcup (\bigcup_{j \in \mathcal{J}} \mathcal{O}_j(t))\}. \quad (44)$$

3) *Action*: In this paper, the action refers to the discrete decision space of UAVs. Both Members and Leaders are required to control the velocity v , where v is determined by the flight direction and speed variation. The flight direction is defined as forward, backward, left, and right, while the speed variation refers to deceleration, constant velocity, and acceleration. Given that a member needs to accomplish target tracking based on results of detection, it will additionally take charge of sensing selection. When a relay link exists, member will make a decision on whether to perform task offloading according to tracking status, so as to achieve load balancing in MLNTN. The action tuples for UAVs are defined as

$$\pi_u(t) = \{\rho_u(t), x_u(t), v_u(t)\}, \quad (45)$$

$$\pi_j(t) = \{v_j(t)\}. \quad (46)$$

4) *Reward*: Reward signals fed back by the environment indicate the effectiveness of the strategy and primarily guide agents in performing updates. We propose a reward shaping method with strengthened relevance, referred to as *joint-process-reward-shaping* (JPRS). It not only facilitates the joint optimization of individual agents in terms of time delay, energy consumption, outage robustness, and resolution preference, but also enhances cooperation of the system in sensing selection, task offloading, and trajectory control. In JPRS, two distinct types of reward functions are employed: task-oriented rewards and phase-oriented rewards. Task-oriented rewards primarily aim to prioritize object detection, requiring the completion of tasks such as image acquisition, detection, tracking, interference avoidance, relay transmission, and resource management, respectively. However, inevitable conflicts arise during task execution. For example, high-resolution images can improve accuracy of detection, assisting in target tracking, while leading to extra resource burdens. To address this challenge, phase-oriented rewards are introduced, which perform a mixed combination of task-oriented rewards according to phases of sensing, communication, computation, and control. Specifically, the determination of coefficient values comprehensively considers the influence between primary and secondary phases, such that under the premise of successfully completing tasks within the primary phase, the benefits provided by the secondary phase are maximized. Notably, the weight coefficient values are set based on experimental experience. In this paper, we define four types of task-oriented rewards, followed by detection reward, tracking reward, outage robustness reward, and resource management reward. For object detection, this process can be defined as

$$\mathbf{r}_{det}(t) = \mu_1 * \eta(t), \quad (47)$$

where the coefficient $\mu_1 = 1$. For target tracking, the task-oriented reward can be defined as

$$\mathbf{r}_{tra}(t) = -D_{t2u} - \mu_2 * I_u(t), \quad (48)$$

where $\mu_2 = 0.5$ is the coefficient. D_{t2u} denotes the sum of the shortest distances between each target and members. For outage robustness, this process can also be written as

$$\mathbf{r}_{rob}(t) = \mu_3 * \Gamma(t) - \mu_4 * I_j(t), \quad (49)$$

where $\mu_3 = 1$, $\mu_4 = 0.25$ are the coefficients. For resource management, the task-oriented reward can be defined as

$$\mathbf{r}_{res}(t) = -\mu_5 * T(t) - \mu_6 * E(t), \quad (50)$$

where $\mu_5 = 0.8$, $\mu_6 = 0.2$ are the coefficients. By weighting task-oriented rewards, phase-oriented rewards are derived as

$$\mathbf{R}_{sen}(t) = \mathbf{r}_{det} + \Omega_1 * \mathbf{r}_{tra} + \Omega_2 * \mathbf{r}_{rob} + \Omega_3 * \mathbf{r}_{res}, \quad (51)$$

$$\mathbf{R}_{com}(t) = \mathbf{r}_{res} + \Omega_1 * \mathbf{r}_{tra} + \Omega_2 * \mathbf{r}_{det} + \Omega_3 * \mathbf{r}_{rob}, \quad (52)$$

$$\mathbf{R}_{con}(t) = \mathbf{r}_{rob} + \Omega_1 * \mathbf{r}_{tra} + \Omega_2 * \mathbf{r}_{det} + \Omega_3 * \mathbf{r}_{res}, \quad (53)$$

where the above represents rewards in sensing, communication as well as computation, and control, respectively. $\Omega_1 = 0.1$, $\Omega_2 = 0.3$, $\Omega_3 = 0.5$ are the coefficients. Therefore, the total reward for MLNTN can be calculated by weighted summation of all phase rewards, and this process can be written as

$$\mathbf{r}_{tot}(t) = \frac{1}{N_{tot}} \left(\sum_u \mathbf{R}_{sen} + \sum_u \sum_j \sum_s \mathbf{R}_{com} + \sum_j \mathbf{R}_{con} \right), \quad (54)$$

where N_{tot} represents the number of UAVs and satellites.

VI. MULTI-AGENT REINFORCEMENT LEARNING SOLUTIONS

A. Proximal Policy Optimization Algorithm

PPO is one of the state-of-the-art RL algorithms that has attained extensive adoption in recent years. Specifically, PPO hinges on a dual-network of actor-critic architecture, consisting of an actor-network $\pi_\theta(a|s)$ parameterized by θ and a critic-network $V_\phi(s)$ parameterized by ϕ . The actor-network will approximate a stochastic policy. In essence, given a representation of state, the actor dictates the likelihood of each action being executed. As for the critic-network, it estimates the state-value function and quantifies the expected cumulative discounted return starting from the current observation. A distinguishing feature of PPO is to introduce a clipped surrogate objective function, thereby curtailing excessive divergence between the new and old policies $\pi_{\theta_{old}}(a|s)$. In the training workflow, the critic updates through minimizing mean squared error (MSE) to enhance the accuracy of return estimation. To reduce variance in return estimation, generalized advantage estimation (GAE) is employed in this paper. Meanwhile, based on estimation of critic, the actor prioritizes actions associated with higher returns. During an episode, $\mathbb{N}$ independent parallel environments are adopted. Each agent interacts with $\mathbb{T}$ time slots in $\mathbb{N}$ independent parallel environments, forming an experience buffer. It is then divided into $\mathbb{M}$ mini batches, each

containing $b = \frac{\mathbb{B}}{M}$ samples, to be used for iteration of loss function, which can be derived as

$$\mathcal{L}_V^{cr}(\phi) = \mathbb{E}_b [(V_t^{tar} - V_\phi(s_t))^2], \quad (55)$$

$$\mathcal{L}_\pi^{ac}(\theta) = \mathbb{E}_b \left[\min \left(r(\theta) \hat{A}_t, \text{clip}(r(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_t \right) \right], \quad (56)$$

$$\text{clip}(a, 1 - \varepsilon, 1 + \varepsilon) = \begin{cases} 1 - \varepsilon, & a < 1 - \varepsilon, \\ 1 + \varepsilon, & a > 1 + \varepsilon, \\ a, & \text{otherwise.} \end{cases} \quad (57)$$

Noting that $\mathcal{L}_V^{cr}(\phi)$ is the loss for the critic, $\mathcal{L}_\pi^{ac}(\theta)$ is the loss for the actor, $r(\theta) = \frac{\pi_\theta(a|s)}{\pi_{\theta_{old}}(a|s)}$, ε is a hyperparameter and V_t^{tar} is the target value. $\hat{A}_t$ is the estimated advantage function, which can be derived as

$$\hat{A}_t = \delta_t + (\gamma\lambda)\delta_{t+1} + \dots + (\gamma\lambda)^{T-t-1}\delta_{T-1}, \quad (58)$$

where $\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$, γ is discount factor and λ is a hyperparameter. However, MLNTN in remote areas involves decision-making by multi-agent, indicating that each agent only gets partial observation instead of complete state. Without loss of generality, the PPO algorithm based on a single central agent cannot be optimal as usual in MAS.

B. PPO under MARL Framework

In the following process, we will take PPO as an example and expand it into the MARL framework based on different frameworks. Under MAS, even if the same action is taken in the same state, the distribution of state transitions and rewards obtained may continue to change. To address this problem, one CTE framework based on fully centralized is proposed, which treats multi-agent as a joint agent making decisions. There is another opposite IL framework based on fully decentralized in MARL. Its main assumption is that each agent learns independently in its own environment, without considering any changes from other agents. So, RL algorithm based on a single agent can be directly applied to each agent. Moreover, the other framework named CTDE is a classic and effective paradigm. During training, the global state combined with the observation information of all agents is used to update the actor-critic network. While testing, inference can be executed in a distributed way by every agent.

It is noteworthy that several techniques such as *sharing parameters*, *normalized GAE* and *Pop-art* are also introduced here to reshape feature vectors or rewards.

1) *Sharing parameters*: In traditional RL, each feature has its own independent parameters. Through sharing parameters, multiple features can share the same parameter in a unified network, promoting convergence. As for homogeneous agents, it can also enhance cooperation with each other and reduce the complexity of interaction. Therefore, considering existence of members and leaders, we only need to define two different types of actor-critic networks. In detail, θ_u and ϕ_u denote member u while θ_j and ϕ_j denote leader j , which are uniformly referred to as θ_{UAV} and ϕ_{UAV} for convenience.

Algorithm 1 MLMAPPO for MLNTN.

Initialize MLNTN environment and actor-critic networks for each type of UAVs (member u and leader j) individually with *orthogonal initialization* method;

- 1: **for** each thread of parallel environments n **do**
- 2: Set replay buffer $\mathbb{B} = \{\}$;
- 3: **for** time slot $t \in \mathbb{T}$ **do**
- 4: **for** each member and leader in UAV swarms **do**
- 5: Update the observations $\mathcal{O}_u(t)$, $\mathcal{O}_j(t)$;
- 6: Input $\mathcal{O}_u(t)$, $\mathcal{O}_j(t)$ after *layer normalization* for feature vectors to the respective actor networks and obtain the actions $\pi_u(t)$, $\pi_j(t)$;
- 7: The critic evaluates based on $\mathcal{S}$ and π after *layer normalization* for feature vectors;
- 8: **end for**
- 9: **for** each leader and satellite **do**
- 10: Allocate computing resource f_j^{max} , f_s^{max} ;
- 11: Calculate the current reward $r(t)$ with JPRS;
- 12: Store $\{s(t), a(t), r(t), s(t+1)\}$ in the buffer $\mathbb{B}$;
- 13: Compute the advantage function A_t after *normalized GAE*, utilizing *Pop-art*;
- 14: Adjust the target value V^{tar} after *Pop-art*;
- 15: **end for**
- 16: **if** $t \% b == 0$ **then**
- 17: Adopt *huber loss* for the critic;
- 18: Compute gradient based on individual loss function and update θ and ϕ while employing *sharing parameters* between homogeneous agents;
- 19: Clear the buffer $\mathbb{B}$ and save model;
- 20: **end if**
- 21: **end for**
- 22: Episode += 1
- 23: **end for**

2) *normalized GAE*: Similarly, apply normalization to the advantage function, i.e., $A_t = (\hat{A}_t - \mu(\hat{A}_t)) / \sigma^2(\hat{A}_t)$, which can maintain the stability of total rewards.

3) *Pop-art*: In every episode of training, the mean and variance of global rewards V^{tar} are preserved through sliding averages initially. Then, the original reward $V_\phi(\cdot)$ is normalized using the mean and variance to update parameters of networks. When varying between the mean and variance, it is necessary to reverse adjust the weights and biases of the network to anti-normalize original output to remain unchanged.

Moreover, compared with mean square error, *huber loss* is more robust to outliers and can be used to update the critic network. Based on this analysis, we can derive the loss function for MAPPO as follows:

$$\mathcal{L}_{MAPPO}^{cr}(\phi_{UAV}) = \mathbb{E}_b \left[\text{huber}_\delta \left(\hat{V}_t^{tar}, V_{\phi_{UAV}}(s_t) \right) \right], \quad (59)$$

$$\mathcal{L}_{MAPPO}^{ac}(\theta_{UAV}) = \mathbb{E}_b \left[\min \left(r(\theta_{UAV}) \hat{A}_t, \text{clip}(\cdot) \hat{A}_t \right) \right], \quad (60)$$

where $\delta = 10$ for huber loss, $\hat{V}_t^{tar}$ is the target value after Pop-art processing, $\hat{A}_t$ denotes the advantage function with normalized GAE and clip($\cdot$) in (57) is designed for $r(\theta_{UAV})$.

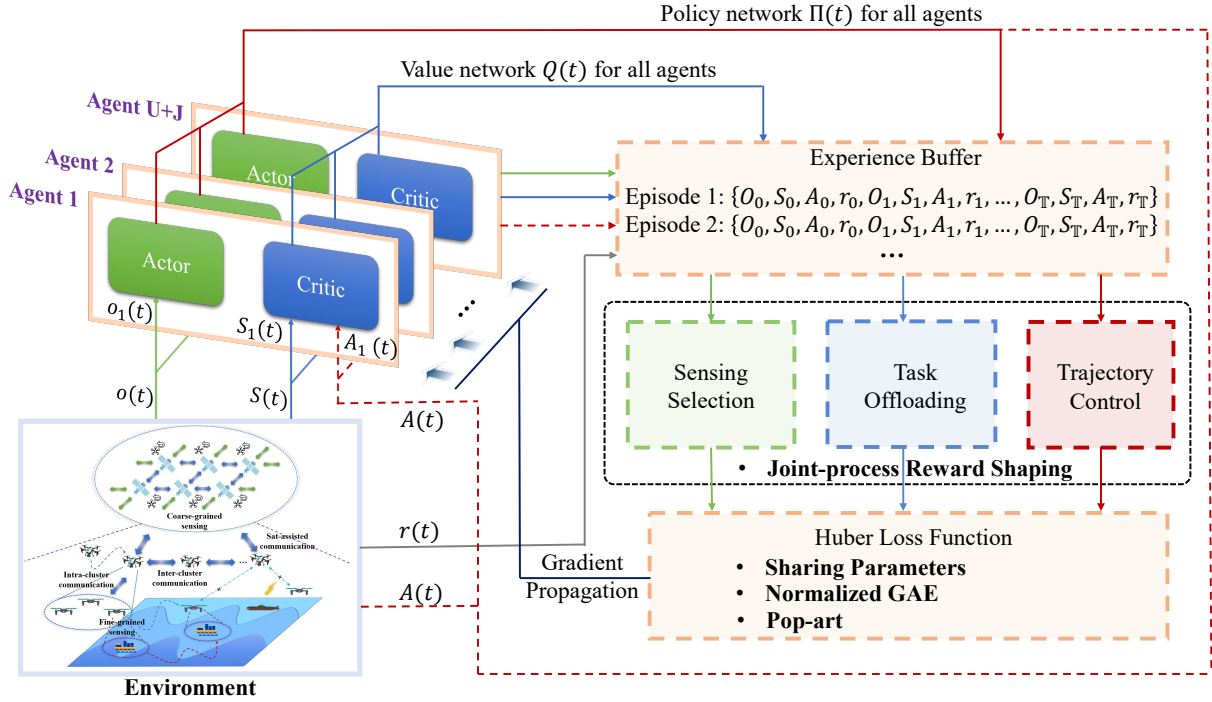


Fig. 6. The workflow of the proposed MLMAPPO algorithm.

Thus, the total loss for MAPPO is defined as

$$\mathcal{L}_{MAPPO} = \mathcal{L}^{cr}(\phi) - c_1 \mathcal{L}^{ac}(\theta) + c_2 \mathcal{S}(\pi_\theta), \quad (61)$$

where ϕ and θ denote for ϕ_{UAV} , θ_{UAV} , respectively. c_1 and c_2 are weight coefficients. $\mathcal{S}$ is an entropy bonus, which can encourage exploration during training.

To address the multi-objective problem, we formally propose *multi-layer multi-agent proximal policy optimization* (MLMAPPO) as well as *multi-layer independent proximal policy optimization* (MLIPPO) algorithms with the above techniques and JPRS for reward shaping. The pseudocode of MLMAPPO is illustrated in Algorithm 1, and the detailed training process is shown in Fig. 6. Specifically, each UAV as an agent interacts with the environment, and updates policies using samples stored in the replay buffer. JPRS is employed to further strengthen the coupling among sensing selection, task offloading, and trajectory control. Several techniques associated with network training are adopted to both promote policy sharing and ensure stable gradient propagation.

As for the MLIPPO, just revise observation of critic to o_t^i for every agent instead of s_t . In particular, since the PPO operates in a centralized framework, there is a need to initially concatenate observations of all agents as a global state o_t and the actions of all agents as a joint action a_t during training. Subsequently, it decouples the joint action a_t into a specific action a_t^i for every agent, which will inevitably cause extra computational complexity. Besides, there is a slight difference between s_t and o_t , which includes an additional ID of agents for s_t . For more professional analysis, we conduct mega experiments in the next section to compare the performance of PPO based on CTE, MLIPPO based on IL, and MLMAPPO based on CTDE.

C. Complexity Analysis

As mentioned in [52], we conduct the analysis of different algorithms in terms of both time and space complexity.

(1) *Time Complexity*: Since the time complexity differs between the training and inference phases, the following discussion is divided into two aspects. On one hand, the time complexity includes both forward and backward propagation during training. For the vanilla PPO, it can be expressed as $\mathcal{O}(\mathbb{T} \cdot (P_\theta + P_\phi)^2)$, where P_θ and P_ϕ denote the number of parameters for the actor and critic networks, respectively. As for MLIPPO, the time complexity can be derived as $\mathcal{O}(\mathbb{T} \cdot N \cdot (P_\theta + P_\phi)^2)$, where N is the number of agents. Since the critic takes states as input, the time complexity of MLMAPPO is $\mathcal{O}(\mathbb{T} \cdot N \cdot (P_\theta + P_\phi)^2 + \mathbb{T} \cdot \mathcal{S})$. On the other hand, during inference, only forward propagation is required. The time complexity of PPO is $\mathcal{O}(P_\theta)$, while both MLIPPO and MLMAPPO have the same time complexity of $\mathcal{O}(N \cdot P_\theta)$.

(2) *Space Complexity*: The space complexity is mainly determined by the size of model parameters and replay buffers. For the vanilla PPO, it can be derived as $\mathcal{O}(\mathbb{B} + P_\theta + P_\phi)$. Accordingly, the space complexity of MLIPPO is given by $\mathcal{O}(\mathbb{B} + N \cdot (P_\theta + P_\phi))$. As for MLMAPPO, since it needs to store the global state in the replay buffer, the space complexity can be expressed as $\mathcal{O}(\mathbb{B} + N \cdot (P_\theta + P_\phi) + \mathcal{S})$.

VII. PERFORMANCE EVALUATION AND ANALYSIS

In this section, the convergence of training is evaluated with different seeds, followed by testing based on multiple rounds for performance analysis.

TABLE I
SIMULATION CONFIGURATION

Notation	Value	Definition
ψ	3.5	Attenuation in space (dB/km)
T_u^{max}	[0.1, 1]	Maximum tolerable latency for tasks (s)
P_k	[5, 10]	Power of jammers (W)
P_u	3	Power of members (W)
P_j	25	Power of leaders (W)
P_s	50	Power of satellites (W)
n	3	Path loss exponent
ϕ	0.8	Efficiency of the antennas
κ	10^{-28}	Energy consumption coefficient
N_0	-174	Noise power density (dBm/Hz)
h^u	2	Antenna height of members (m)
h^j	5	Antenna height of leaders (m)
B_u, B_j	5	Channel bandwidth of UAVs (MHz)
B_s	10	Channel bandwidth of satellites (MHz)
f_j^{max}	1	CPU frequency at leaders (GHz)
f_s^{max}	10	CPU frequency at satellites (GHz)
v_s	7500	Velocity of satellites (m/s)
α_s	20	Azimuth angle ($^\circ$)
d_{min}	200	Minimum breakpoint (m)
d_{max}	400	Maximum breakpoint (m)
Ω_j	0.3	Antenna radius of leaders (m)
Ω_s	1	Antenna radius of satellites (m)
R	6371	Radius of the Earth (km)
d_{L2L}	1	Reference distance for L2L link (km)
d_{L2S}	10	Reference distance for L2S link (km)

A. Simulation Configuration

Specifically, we consider a limited horizontal region of 1 km * 1 km for MLNTN, with 3 targets and 2 malicious jammers. The set of optional image resolutions is $G = \{360p, 540p, 720p, 900p\}$. There are 3 UAV swarms in total, composed of 4 members at an altitude z_u of 100 m and 3 leaders at an altitude z_j of 500 m. Each UAV swarm has only one leader. The detection field for members is a circular region with a radius of $R^{det} = 200$ m, and the routing area for leaders is also a circular region with a radius of $R^{rob} = 350$ m. In terms of computational properties for object detection, the data size of acquired images by members and satellites is distributed in [50, 100] KB and [10, 50] KB, respectively. While the computing density for members and satellites is [1500, 2000] CPU cycles/bit and [1000, 1500] CPU cycles/bit, respectively. The LEO satellite constellations at an altitude z_s of 500 km adopt Walker Delta mode with $45^\circ: 50/5/1$ setting, where 45° denotes the inclination angle, 50 is the total number of satellites, 10 is the number of satellites per orbit, and 1 is the parameter of inter-plane spacing. It is noted that each leader in a UAV swarm can communicate with a specific satellite, which means actually 3 satellites are always in service. Regarding UAVs, there are limitations for maximum flying speed $v_u^{max} = 5$ m/s and $v_j^{max} = 5$ m/s. The CPU frequency at members is fixed at 0.1 GHz, while the assignable maximum CPU frequency for leaders and satellites is 1 GHz and 10 GHz, respectively. The carrier frequency f_{M2L} , f_{L2L} , and f_{L2S} are all set to 2.4 GHz. Detailed illustrations of definitions are summarized in Table I.

Considering the parameters of MLMAPPO and MLIPPO, both the actor and critic networks comprise two fully connected hidden layers with 64 neurons. Every episode has a length $\mathbb{T}$ of 300 steps. We define the weights of the penalty

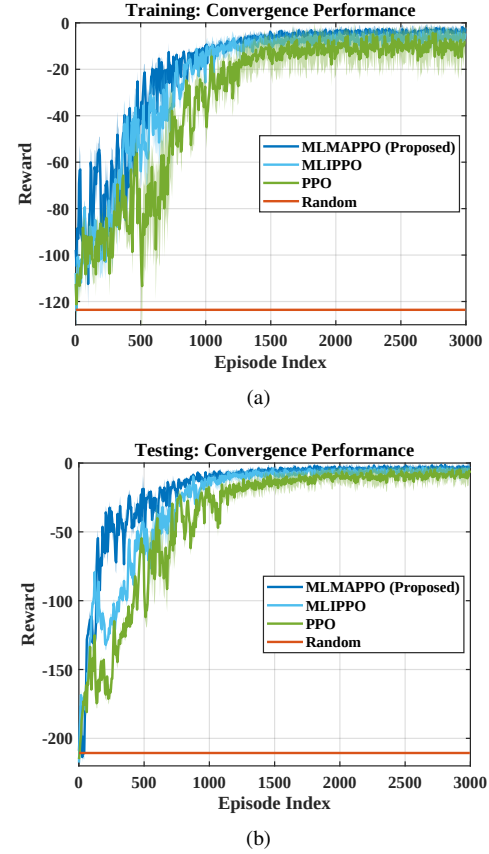


Fig. 7. Total rewards versus the number of episodes during training and testing. Notably, the solid line indicates the mean value of rewards, while the shaded area represents the variance value of rewards across different seeds. (a) Training results. (b) Testing results.

terms $\kappa^c = 0.1$ in reward. The learning rate for actor and critic networks is set to $5e-5$. The replay buffer size and batch size are defined as 2048 and 64, respectively. In the training and testing, we keep the number of episodes to 3000 equally.

B. Convergence Evaluation

Fig. 7 shows the convergence of total reward for MLNTN in training and testing. The current solutions for comparison include MLMAPPO, MLIPPO, PPO, and Random algorithms. Noting that all UAVs are randomly initialized with flight coordinates throughout the entire range in training, while initialized with flight coordinates from the same starting-point range during testing. Furthermore, we firstly train the actor-critic network based on every algorithm to get well-trained models. Then, the well-trained models are used for evaluating convergence and performance in a new environment when testing, making it more practical in the real world.

For each algorithm to be analyzed, we run different seeds for training and calculated the mean and variance value of the results, while using the solid lines and the shaded areas to represent them in figures, respectively. Overall, all algorithms can realize a relatively stable convergence curve in 3000 episodes we set. Here, Fig. 7a denotes the results of convergence in training and Fig. 7b is the results in testing. As described in these figures, the MARL algorithms have achieved better

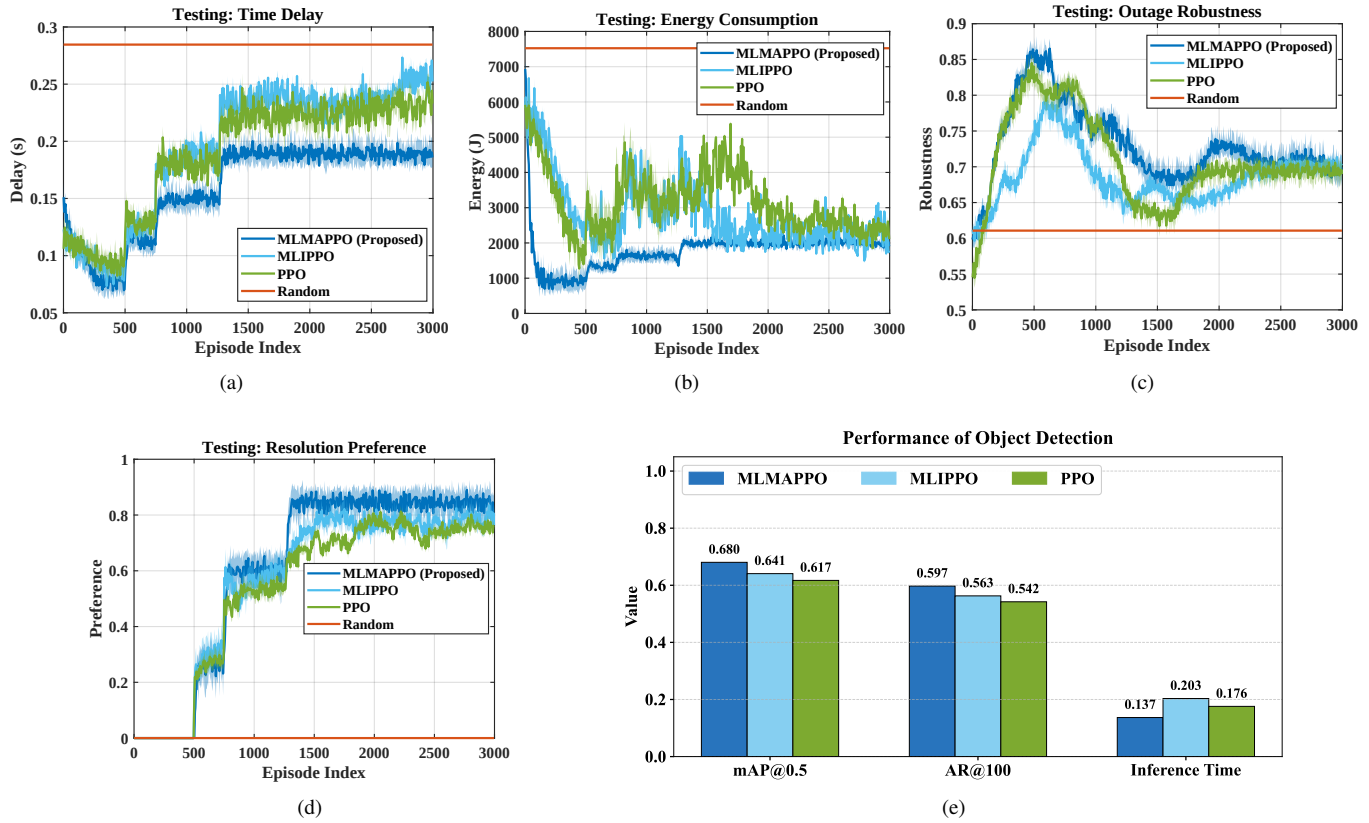


Fig. 8. Performance of MLMAPPO, MLIPPO, PPO and Random algorithms across various quantitative indicators in testing. (a) Time delay. (b) Energy consumption. (c) Outage robustness. (d) Resolution preference. (e) Detection metrics.

results than Random. However, there is still a gap between the PPO and the MLIPPO algorithm based on the IL framework. One possible explanation for the PPO algorithm is that the larger action space, invoked by stacking all actions, brings an extra challenge when decoupling. As for the IPPO, given selfish learning, each agent naturally prioritizes maximizing its own reward rather than considering the global optimum. Compared to them, it is obvious that our proposed MLMAPPO algorithm based on the CTDE framework outperforms other baselines whether it's training or testing, with the highest mean value and relatively small variance.

C. Performance Analysis

As shown in Fig. 8, the performance of various indicators is analyzed in testing, involving time delay, energy consumption, outage robustness, resolution preference and detection metrics. Here, time delay and energy consumption are common variables in extensive research, which mainly indicate the immediacy of task completion and resource management. Regarding outage robustness represented by resilience, it considers outage of communication, including interference from jammers and outage of transmission links. In addition, since the accuracy of detection increases with the resolution of the image in a certain range, we define resolution preference as the proportion of acquiring fine-grained images. To describe the improvement in detection, we introduce mean average precision (mAP), average recall (AR) and inference time. In particular, all targets

did not enter the detection field of members initially. Object detection can be assisted for targets tracking only by acquiring coarse-grained images through satellites at this moment. It is worth noting that we discover the optimal strategy for object detection is to perform sensing and computation directly on satellites during the initial phase.

In Fig. 8a, MLMAPPO exhibits better performance, showing a 22.15% reduction in time delay compared to PPO. The overall trend of time delay presents a pattern of initial reduction followed by a gradual increase. During the early stages of testing, satellite constellations undertake sensing and computing at this period. As UAVs approach targets, MLNTN prefers fine-grained images. Considering that image resolution determines the computational complexity, whenever the images for detection are acquired by UAVs, the time delay with a relatively high weight (i.e., $\mu_5 = 0.8$) exhibits a pronounced change. Since a total of three targets are considered, three distinct abrupt changes are observed, resulting in a step-like variation pattern.

When it comes to energy consumption, its changing is similar to Fig. 8a as shown in Fig. 8b. It is evident that the PPO and the MLIPPO algorithm exhibit significant fluctuations, while the MLMAPPO keeps a steady trend and shows a 9.67% reduction compared to PPO. Because the value of energy consumption is relatively large, it manifests a pronounced trend of rapid decline when agents make correct decisions. Subsequent to sensing targets by members for high accuracy

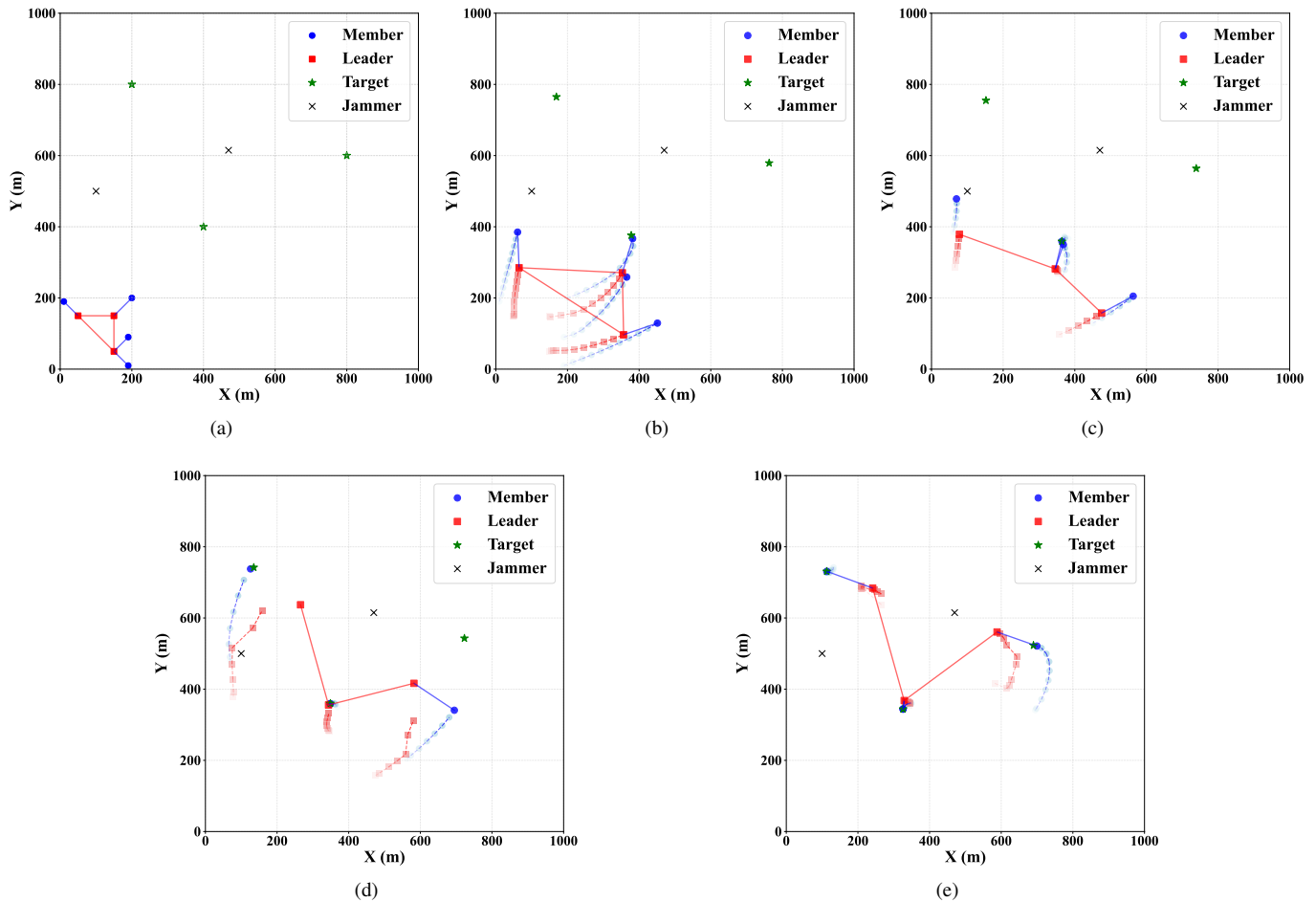


Fig. 9. Circle points and dashed lines represent historical positions and trajectories, respectively. The solid lines represent the communication links between UAVs, including M2L by blue and L2L by red. Sub-figure (a) indicates initial positions and (b), (c), (d), (e) correspond to the moments when members track all targets sequentially from start to end, among which (b) and (c) track the same target.

of detection, the computation of fine-grained images further leads to an increase in energy. So, the energy consumption also exhibits a gradual elevation in a total of three stages. Interestingly, we have observed that when all members opt to acquire fine-grained images, offloading the detection tasks to leaders for edge computing emerges as the optimal strategy.

We utilize the total number of available links established in MLNTN to represent outage robustness, as illustrated in Fig. 8c. On the whole, MLMAPPO outperforms the vanilla PPO with a more stable variation and a slight 1.88% gap. More specifically, since UAVs depart from the same initial location, a number of links are naturally established, leading to a rapid increase in outage robustness during the early stages of testing. Subsequently, since targets are far apart from one another, joint tracking becomes infeasible, which forces members to perform distributed tracking and inevitably reduces the outage robustness. Moreover, as UAVs approach targets, they may suffer from malicious interference from jammers, which further damages link establishment and results in persistent fluctuations. Ultimately, the outage robustness exhibits an overall rise-then-fall trend with fluctuations.

Resolution preference serves as a critical factor describing the performance of collaborative sensing for heterogeneous

units, which is illustrated in Fig. 8d. It can be concluded that MLMAPPO achieves the highest proportion of fine-grained images, showing a 13.84% improvement compared to PPO. In the experiments, a relatively high weight is assigned to the resolution preference (i.e., $\mu_1 = 1$), encouraging MLNTN to favor images acquired by the UAV. This is because high-resolution images are beneficial for improving detection accuracy, thereby facilitating target tracking. Consequently, whenever the images for detection are acquired by the UAV, the resolution preference generates a sudden reward increase to encourage such a strategy. Ultimately, a step-like variation pattern similar to that of the time delay is formed.

As for detector-level performance, we mainly analyze the mAP@0.5, AR@100 and inference time under different algorithms in Fig. 8e. It is observed that MLMAPPO achieves the best detection performance with the highest mAP@0.5 and AR@100, with improvements of 10.21% and 10.14% over PPO, respectively. However, MLIPPO achieves a slightly longer inference time than PPO, meaning that independent policies can hardly adapt to the available edge computing resources in a timely manner. In contrast, MLMAPPO achieves the lowest inference time, which further verifies its superiority for object detection.

To specifically demonstrate the trajectory control process of UAV swarms, we have captured key snapshots from the experiments under the MLMAPPO algorithm in Fig. 9. It is further verified that members and leaders can not only accomplish specific tasks, but also achieve cooperation.

As shown in Fig. 9a, it describes the initial positions at a corner. There are three UAV swarms responsible for tracking the same number of targets, and one of the swarms has two members. Fig. 9b presents a key snapshot captured at the moment when the first target is successfully tracked. It is noteworthy that since the first target is relatively close, the extra member is assigned to the corresponding swarm. Thus, the first target is repeatedly tracked in Fig. 9c. Meanwhile, one member, being in proximity to the jammer, causes interference outage at this point, resulting in the disappearance of the M2L link represented by the blue solid line. Subsequently, the members and leaders timely evaded the jammer and successfully tracked the second target in Fig. 9d. However, due to the requirement for timeliness in tracking, the distance between the member and the leader became relatively large, which once again leads to the outage of the M2L link, impairing the reliability and resilience of MLNTN. Later, the leaders adjust their relative positions with respect to each other and establish multi-hop relaying with all members. Under relaying of the routing network, the last target is successfully tracked by the members in Fig. 9e. To summarize, with the aid of assisted offloading by leaders and satellites, UAV swarms have learned advanced trajectory control strategies and successfully achieved distributed tracking based on object detection.

VIII. CONCLUSION AND FUTURE WORK

In this paper, we have investigated object detection for tracking. We have designed a hierarchical architecture named ML-NTN, with clusters as operational units. Several crucial factors related to detection and tracking for clusters have been modeled, involving malicious interference, communication channels, energy consumption, and outage robustness. Furthermore, we have proposed a collaborative object detection approach with resolution preference to address heterogeneous sensing in NTN. To support distributed strategies, we have proposed two MARL methods, named MLMAPPO and MLIPPO respectively, jointly optimizing sensing selection, task offloading, and trajectory control as a multi-objective problem. Simulation results demonstrate that MLMAPPO achieves a more effective balance between execution timeliness and network resilience.

As part of our future work, we plan to investigate OEC under federated learning (FL) or split learning (SL). Specifically, our goal is to explore the scheduling problem on a global scale to achieve service balance. In addition, mega constellations possess distinct topology structures, and introducing graph neural networks (GNNs) from graph learning (GL) for feature extraction remains a promising direction.

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